



COMPETENCY-FOCUSED PRACTICE QUESTIONS

ICSE – CLASS X

**ENVIRONMENTAL
SCIENCE**

PREFACE

With a growing emphasis on competency-based education globally, the educational landscape in India has also steered towards high-quality learning experiences that allow learners to incorporate critical thinking and problem-solving approaches. This approach goes beyond rote memorisation and focuses on developing the skills and knowledge that students need to apply in their real-world scenarios.

The Council for the Indian School Certificate Examinations (CISCE), as a national-level progressive examination board, has taken several steps to infuse competency-based education in CISCE schools through teacher capacity-building on item development for competency-based assessments and the incorporation of competency-focused questions at the ICSE and ISC levels from the examination year 2024.

To further facilitate the adoption of competency-based assessment practices in schools and to support teachers and students towards the preparation for attempting higher-order thinking questions in future board examinations, Item Banks of **Competency-Focused Practice Questions** for selected subjects at the ICSE and ISC levels have been developed. This Item Bank consists of a rich variety of questions, both objective and subjective in categories, aimed at enhancing the subject-specific critical and analytical thinking skills of the students.

In this Item Bank, each question is accompanied by the topic and cognitive learning domain/s that it intends to capture. The cognitive domains reflected in these questions include understanding, analysis, application, evaluation and creativity, along with some questions of the higher-order recall domain. The Answer Key at the end presents the possible answers to a given question, but it is neither limiting nor exhaustive.

These practice questions are also meant to serve as teacher resources for classroom assignments and as samplers to develop their own repository of competency-focused questions. Apart from offering a good practice of higher-order thinking skills, engaging with these questions would allow students to gauge their own subject competencies and use these *assessments for learning* to develop individual learning pathways.

During the development of this Item Bank, a large pool of questions was prepared by a team of experienced CISCE teachers. The questions that were finalised by the internal and external reviewers as being higher-order competency-focused questions have been collated in this item bank.

I acknowledge and appreciate all the ICSE and the ISC subject matter experts who have contributed to the development and review of these high-quality competency-focused questions for CISCE students.

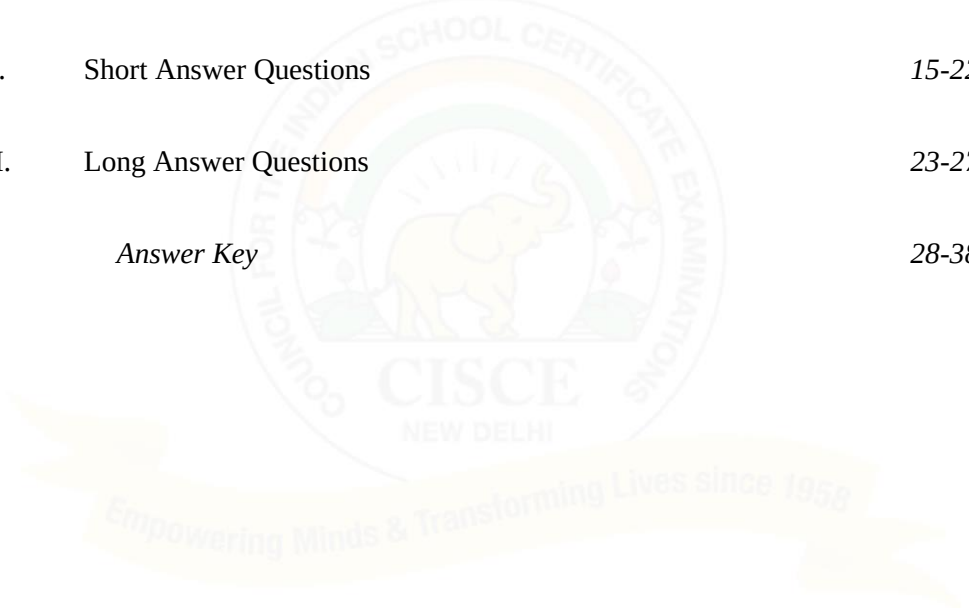
We are hopeful that teachers and students will utilise these questions to support their teaching-learning processes.

August 2024

Dr. Joseph Emmanuel
Chief Executive & Secretary
CISCE

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COMPETENCY-FOCUSED PRACTICE QUESTIONS

ICSE - CLASS X

Environmental Science

I: Multiple Choice Questions (MCQ) (1 Mark)

S.No.	Questions
1.	<i>[Towards a Sustainable Future]</i> Intergenerational equity involves: (a) minimising the risk to future generations. (b) reducing financial risk to future generations. (c) maximising freedom of choice. (d) all of the above. [Analysis]
2.	<i>[Environment and Development]</i> Which of the following is INCORRECT in the case of fair trade? (a) They create equal opportunities for economically disadvantaged producers. (b) They help in the creation of social projects like education and housing (c) Both (a) & (b). (d) None of the above. [Evaluate]
3.	<i>[Towards a Sustainable Future]</i> Remote sensing satellites include: (a) observation by the naked eye. (b) photography by a camera. (c) photography from aircraft. (d) sensing by sensors from space satellites. [Evaluate]
4.	<i>[Environment and Development]</i> Assertion (A): Environmental ethics is the application of ethical study to environmental concerns Reason (R): A key aspect of environmental ethics is developing and justifying the theory of the moral relations between humans, non-human species and the natural environment. (a) Both (A) and (R) are true, and (R) is the correct explanation for (A). (b) Both (A) and (R) are true, but (R) is not the correct explanation for (A). (c) (A) is true, and (R) is false. (d) Both the (A) and (R) are false. [Recall]

S.No.	Questions										
5.	<i>[Waste]</i> Name the process of application of biotechnology for the treatment of waste. (a) Eutrophication (b) Mineralisation (c) Carbonification (d) Bioremediation [Recall]										
6.	<i>[Waste]</i> Match the following: <table border="0" style="width: 100%;"> <thead> <tr> <th style="text-align: left;">Column A</th><th style="text-align: left;">Column B</th></tr> </thead> <tbody> <tr> <td>I. Montreal Protocol</td><td>A. Reduction in Volatile organic compounds</td></tr> <tr> <td>II. Kyoto Protocol</td><td>B. Reduction of CFCs</td></tr> <tr> <td>III. Geneva Protocol</td><td>C. Reduction in greenhouse gases</td></tr> <tr> <td>IV. Earth Summit</td><td>D. Sustainable development</td></tr> </tbody> </table> (a) I-B, II-C, III-A, IV-D (b) I-C, II-B, III-D, IV-A (c) I-B, II-C, III-D, IV-A (d) I-B, II-A, III-D, IV-C [Analysis]	Column A	Column B	I. Montreal Protocol	A. Reduction in Volatile organic compounds	II. Kyoto Protocol	B. Reduction of CFCs	III. Geneva Protocol	C. Reduction in greenhouse gases	IV. Earth Summit	D. Sustainable development
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I. Montreal Protocol	A. Reduction in Volatile organic compounds										
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7.	<i>[Biodiversity]</i> <i>Some international environmental agreements are themselves trade agreements.</i> Which of the following best supports the above statement? (a) Convention on International Trade in Endangered Species (CITES) (b) Earth Summit (c) Biodiversity Act (d) Wildlife Protection Act [Application]										

S.No.	Questions
8.	<i>[Towards A Sustainable Future]</i> Country with rich and poor  Which of the following is reflected in the picture given above? (a) Intergenerational (b) Intragenerational (c) Septuagenarians (d) None of the above. [Analysis]
9.	<i>[Food]</i> The increasing demand for food by the growing population leads to: (a) conversion of forest into expanding infrastructure. (b) conversion of forest into agricultural land. (c) migration of rural people into cities. (d) Escalation of the use of fossil fuels. [Application]
10.	<i>[Addressing Population]</i> Women-centred development does NOT focus on: (a) improving girl child's education. (b) empowering women to earn and have control over resources. (c) dismantling the caste system. (d) reducing violence against women. [Analysis]
11.	<i>[Food]</i> Which of these is an advantage of HYV seeds? (a) Yields more under proper irrigation. (b) Higher doses of fertilisers are effective. (c) Easily attacked by pests. (d) Gives better yield of crops. [Analysis]

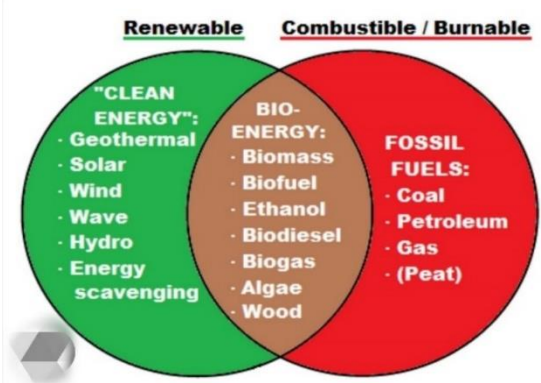
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12.	<i>[Food]</i> Biofertilisers are composed of: (a) crop residue after harvesting. (b) animal dung. (c) living microorganisms like bacteria and protozoa. (d) earthworm castings. [Recall]
13.	<i>[Food]</i> Statement A: Presently, chemical fertilisers are more popular as compared to organic manure. Statement B: Chemical fertilisers are easier to apply, cost-effective and contain three main elements: phosphorus, potassium & nitrogen. (a) Both Statements A & B are incorrect. (b) Statement A is correct, and Statement B is incorrect. (c) Statement A is correct, and Statement B is the reason for A. (d) Statement A is correct, and Statement B is not the reason for A. [Application]
14.	<i>[Food]</i> <i>Conservation tillage improves soil health and limits the extent of soil erosion.</i> Which of the following criteria makes conservation more important? (a) Reduces runoff. (b) The severity of some diseases can increase. (c) Perennial weeds may appear. (d) Pest attack increases. [Analysis]

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15.	<i>[Food]</i> Match each item in Column I with one item in Column II and choose the correct answer from the options below: <table> <tr> <th>Column I</th><th>Column II</th></tr> <tr> <td>A. Dry farming</td><td>I. Contour farming</td></tr> <tr> <td>B. Cutting furrows into the slope of land.</td><td>II. Strip cropping</td></tr> <tr> <td>C. Alternate rows of heavy-rooted & loose-rooted plants.</td><td>III. Tree cropping</td></tr> <tr> <td>D. Planting seedlings over deforested areas.</td><td>IV. Drought intensive crops</td></tr> </table> (a) A-I, B-II, C-III, D-IV (b) A-II, B-III, C-IV, D-I (c) A-IV, B-I, C-II, D-III (d) A-III, B-I, C-II, D-IV [Analysis]	Column I	Column II	A. Dry farming	I. Contour farming	B. Cutting furrows into the slope of land.	II. Strip cropping	C. Alternate rows of heavy-rooted & loose-rooted plants.	III. Tree cropping	D. Planting seedlings over deforested areas.	IV. Drought intensive crops
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16.	<i>[Addressing Population]</i> Around 25 large trees by the side of a water body in the Nalban fisheries area near Salt Lake, Kolkata, have allegedly been chopped down. <i>(Source: The Times of India, Kolkata, 19 February 2024.)</i> The above incident may have been caused by: (a) poor land management. (b) the enormous increase in population. (c) the roots of the trees become weak due to soil erosion. (d) termite infestation at the roots of the trees. [Application]										

S.No.	Questions
17.	<i>[Food]</i> Based on the above figure, sustainable agriculture is: - ensuring the economic viability of farm operations only. - satisfying human demands for food and fibre mainly. - protecting the environment. - meeting the demands of the present generation without compromising the needs of the future generations. [Analysis]
18.	<i>[Energy]</i> The alternate energy that can be used directly from the source is: - solar energy. - wind energy. - geothermal energy. - biomass energy. [Application]
19.	<i>[Biodiversity]</i> Wildlife sanctuaries are NOT: - habitat centric. - species centric. - under private land ownership. - accessible to human activities. [Analysis]
20.	<i>[Energy]</i> Gasohol is an alternative fuel consisting of a mixture of: 90% gasoline & 10% ethanol. 90% gasoline & 10% methanol. 10% gasoline & 90% methanol. 10% gasoline & 90% ethanol. [Recall]

S.No.	Questions
21.	<i>[Biodiversity]</i> <i>In situ</i> conservation does NOT allow: (a) protection of indigenous species in natural habitats. (b) captive breeding. (c) natural selection and community evolution. (d) better conservation on a long-term period. [Analysis]
22.	<i>[Energy]</i> <i>To curb pollution, Delhi bus transport system has become the biggest CNG-run transport system in the world.</i> The main reason for using CNG is: (a) CNG is easily available (b) CNG is more efficient, & less polluting (c) CNG fuel tanks are bulky (d) CNG gives less mileage. [Understanding]
23.	<i>[Energy]</i> Statement A: Wind energy is a renewable source of energy, free of greenhouse gases but not reliable. Statement B: The amount of electricity produced by wind energy varies with wind velocity between zero and storm velocity. (a) Both Statement A & Statement B are incorrect. (b) Statement A is correct & Statement B is the correct explanation of Statement A. (c) Statement A is correct, but Statement B is not the correct explanation of Statement A. (d) Statement A is correct & Statement B is incorrect. [Analysis]

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24.	<i>[Energy; Biodiversity]</i> Match each item in Column I with one item in Column II and choose the correct answer from the options given below: <table> <tr> <th>Column I</th><th>Column II</th></tr> <tr> <td>A. Gene Banks</td><td>I. Red list of threatened species</td></tr> <tr> <td>B. IUCN</td><td>II. Biomass energy</td></tr> <tr> <td>C. Nuclear Energy</td><td>III. Prevents genetic erosion</td></tr> <tr> <td>D. Energy Plantation</td><td>IV. A significant quantity of electricity</td></tr> </table> (a) A-III, B-I, C-IV, D-II (b) A-II, B-III, C-I, D-IV (c) A-IV, B-II, C-III, D-I (d) A-I, B-III, C-IV, D-II [Application]	Column I	Column II	A. Gene Banks	I. Red list of threatened species	B. IUCN	II. Biomass energy	C. Nuclear Energy	III. Prevents genetic erosion	D. Energy Plantation	IV. A significant quantity of electricity
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25.	<i>[Biodiversity]</i> Alpine plants may alter height or leaf size to cope with early snow melt conditions. (Source: MONGABAY, by T. V. Padma, 4th January 2024) What could be the MOST plausible cause? (a) Land degradation (b) Global warming (c) Desertification (d) Intensive agriculture [Analysis]										

S.No.	Questions
26.	<i>[Energy]</i>  With reference to the above Venn diagram, name a renewable bio-energy source that is combustible. (a) Biogas (b) Wave energy (c) Coal (d) Petroleum [Recall & Application]
27.	<i>[Managing Soil and Land]</i> The aim of land reforms is: (a) escalating poverty. (b) increasing agricultural productivity. (c) declining co-operative farming. (d) encouraging land fragmentation. [Application]
28.	<i>[Managing Soil and Land]</i> Terrace farming contributes to sustainable agriculture by: (a) promoting monoculture. (b) maximising landslides. (c) reducing wastage of irrigation water. (d) increasing surface runoff. [Recall]

S.No.	Questions
29.	<i>[Wastage]</i> The long-term impact of uncontrolled e-waste on the environment are: I. loss of valuable raw materials II. decreasing carbon print. III. contamination of groundwater. IV. safeguarding the habitats. Choose the correct options: (a) I and III. (b) II and IV (c) I and IV (d) II and III <i>[Analysis]</i>
30.	<i>[Wastage]</i> Ram performed a waste audit in his local community and found that it could pose a risk to human health. Which of the following four tenets (4R's) did he implement to manage the waste? (a) Rewrite, Reuse, Reduce, Rewarm (b) Rejoice, Reduce, Rewrite, Reuse (c) Rewarm, Recycle, Refuse, Rejoice (d) Refuse, Reuse, Recycle, Reduce <i>[Understanding]</i>
31.	<i>[Managing Soil and Land]</i> Assertion (A): Windbreaks usually consist of alternate rows of trees and shrubs. Reason (R): Windbreaks around a field can raise the cost of heating and cooling and save energy. (a) A is true, and R is false. (b) A is true, and R is false. (c) Both A and R are true, and R is the correct explanation of A. (d) Both A and R are true, and A is the correct explanation of R. <i>[Analysis]</i>
32.	<i>[Managing Soil and Land]</i> <i>Integrated Rural Development Programme is a self-employment programme of the Government of India to raise the income generation capacity of the poor.</i> Which scheme is undertaken under this programme? (a) Urban Sanitation Programme. (b) International Food for Work Programme. (c) Training of pensioners for self-employment. (d) Development of Women and Children in Rural areas. <i>[Recall & Analysis]</i>

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33.	<i>[Wastage]</i> Dengue cases in India have reached nearly 95,000 between January and July 31, 2023, with numbers spiking after excessive rains caused water logging. (Source: <i>The Economic Times</i>, Sept 27, 2023) Water logging occurs because of: (a) proper public sanitation. (b) over-irrigation of fields. (c) adequate drainage. (d) soil impermeability. [Application]										
34.	<i>[Managing Soil and Land]</i> Match each item in Column I with one item in Column II and choose the correct answer from the code given below: <table style="width: 100%; border: none;"> <thead> <tr> <th style="text-align: left;">Column I</th><th style="text-align: left;">Column II</th></tr> </thead> <tbody> <tr> <td>A. Terracing</td><td>(i) Furrows</td></tr> <tr> <td>B. Contour Ploughing</td><td>(ii) Reforestation</td></tr> <tr> <td>C. Bunds</td><td>(iii) Stepped platforms</td></tr> <tr> <td>D. Tree planting</td><td>(iv) Embankments</td></tr> </tbody> </table> (a) A-(i), B-(iii), C-(ii), D-(iv) (b) A -(iii), B-(ii), C-(iv), D-(i) (c) A-(ii), B-(iv), C-(i), D-(iii) (d) A-(iii), B-(i), C-(iv), D-(ii) [Analysis]	Column I	Column II	A. Terracing	(i) Furrows	B. Contour Ploughing	(ii) Reforestation	C. Bunds	(iii) Stepped platforms	D. Tree planting	(iv) Embankments
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S.No.	Questions
35.	<i>[Managing Soil and Land]</i> <pre> graph TD A((Controlled forest grazing)) --- B((Environmental Conservation)) A --- C((Sustainable grazing practice)) A --- D((Monitoring grazing practices)) A --- E((Protection of sensitive areas)) </pre> Which of the following BEST fits the figure above on controlled forest grazing? (a) National Sustainable Policy, 1968. (b) National Grazing Policy, 1978 (c) National Forest Policy, 1988. (d) National Protection Policy, 1998. [Application]
36.	<i>[Managing the Urban Environment]</i> What is the purpose of constructing green buildings in urban areas? (a) To create awareness about money matters in the urban poor. (b) To use less water and renewable sources of energy. (c) To reduce the indoor temperature of rooms. (d) To provide job opportunities for urban youth. [Analysis]
37.	<i>[Controlling Air Pollution]</i> Assertion (A): We lose productive time due to traffic jams on roads. Reason(R): Increasing the number of public transportation in cities can reduce traffic congestion. (a) A is true, and R is false. (b) A is false, and R is true. (c) Both A and R are true. (d) Both A and r are false. [Analysis]

S.No.	Questions
38.	<i>[Managing the Urban Environment]</i> Which of the following is NOT a component of the urban environment? (a) Waste management (b) Transportation planning (c) Agricultural development (d) Water Resources Management [Recall]
39.	<i>[Managing the Urban Environment]</i>  The above picture refers to: (a) a zone where maximum urban development occurs. (b) areas in cities where houses are provided by the government. (c) unplanned growth of urban areas. (d) preservation of green spaces within cities. [Application]
40.	<i>[Controlling the Air Pollution]</i> Statement I: Battery-operated vehicles reduce air pollution Statement II: Battery-operated vehicles are popular in urban areas. Which of the statements is true? (a) Statements I and II are true. (b) Statement I is true, and Statement II is the correct reason for I. (c) Statement I is false, and Statement II is correct. (d) Statement I is true, and Statement II is the correct reason for I. [Analysis]
41.	<i>[Managing the Urban Environment]</i> Which approach is more beneficial to develop existing urban areas? (a) Encroaching forest area near the city. (b) Redeveloping slums and old buildings to accommodate more houses. (c) Providing civic amenities to slums. (d) Converting agricultural lands to residential areas. [Analysis]

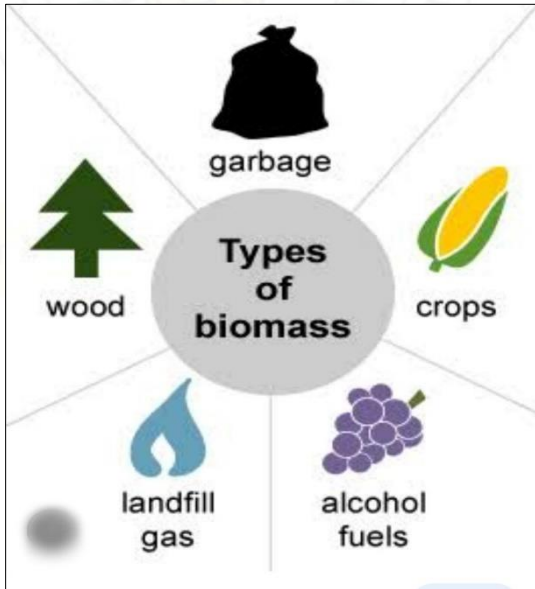
S.No.	Questions
42.	<i>[Controlling the Air Pollution]</i> Which one of the following helps to reduce urban vehicular pollution? (a) Encouraging carpooling and use of public transport. (b) Replacing kerosene with LPG. (c) Using electrostatic precipitators in industries. (d) Encouraging community forestry. [Recall]
43.	<i>[Controlling the Air Pollution]</i> Mr. X bought a motorbike with a <i>four-stroke engine</i>. Mr. Y bought a motorbike with a <i>two-stroke engine</i>. Mr. Z bought a car with a <i>catalytic converter</i>. Name the persons who are likely to contribute less to air pollution. (a) Only X. (b) Only Y. (c) X and Z. (d) Z and Y. [Application]

II: Short Answer Questions (2 Marks)

S.No.	Questions
44.	<i>[Food]</i> “One of every six people in developing countries is not getting enough food to eat.” State <i>any two</i> reasons for the statement above. [Understanding]
45.	<i>[Environment and Development]</i> <i>International aid from agencies has been flowing from developed countries to developing countries.</i> Name <i>any two</i> international aid-giving agencies. [Recall]
46.	<i>[Towards a Sustainable Future]</i> State <i>any two</i> ways of supporting developing countries to follow the Montreal Protocol. [Recall & Understanding]

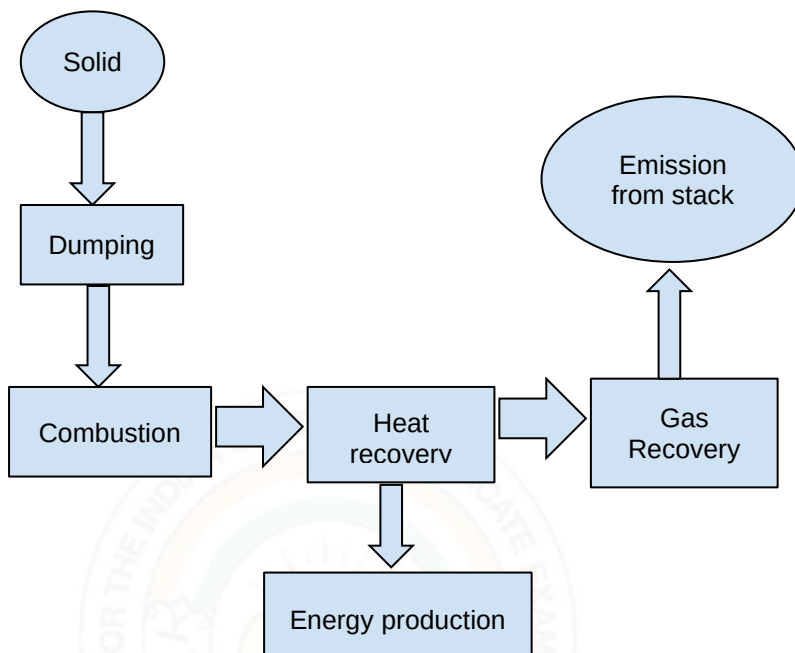
S.No.	Questions
47.	<i>[Towards a Sustainable Future]</i> Figures (a) and (b) illustrate the changes in the shape of the age-sex pyramid for India. By observing the figure, answer the following question: The population planning policy has succeeded only marginally. State whether the statement is true or false. Give an appropriate reason for your answer. [Analysis]
48.	<i>[Towards a Sustainable Future]</i> Chinese manufacturers discovered a market for their plastic toys in India, where the price point for toys is higher. With their creative designs and lower pricing, these plastic toys made in China have now become more popular in Indian markets. Regarding the information given above, is it an example of an unfair trade? Why? [Understanding & Analysis]
49.	<i>[Towards a Sustainable Future]</i> Suggest any two methods of growing food by copying nature. [Understanding]
50.	<i>[Biodiversity]</i> List two ways in which human lifestyle directly degrades wetlands. [Recall]
51.	<i>[Food]</i> Name a pest control method that uses predator insects. Give an advantage of it. [Recall & Understanding]

S.No.	Questions
52.	<i>[Food]</i> Mention two constraints faced by the farmers in the implementation of integrated plant nutrient programmes. [Understanding & Recall]
53.	<i>[Food]</i> What role do gene banks play in biodiversity conservation? Give an example of a gene bank. [Understanding & Recall]
54.	<i>[Addressing Population]</i> Analyse the role of small-scale industries in providing employment opportunities in rural and urban areas. How does this contribute to poverty alleviation? [Understanding & Analysis]
55.	<i>[Food]</i>  Mention two advantages of this device used in sustainable agriculture. [Recall]
56.	<i>[Biodiversity]</i> Mention the significance of the core & buffer zone of the Nilgiri Biosphere Reserve. [Understanding]
57.	<i>[Biodiversity]</i> Imagine you are a wildlife conservationist tasked with protecting endangered species. Analyse the specific challenges faced by critically endangered species compared to vulnerable species. [Analysis]
58.	<i>[Biodiversity]</i> The Convention on Biological Diversity is an agreement adopted at the Earth Summit, Rio de Janeiro, in 1992. Mention two of its main objectives. [Recall]

S.No.	Questions
59.	<i>[Biodiversity]</i>  List two steps that were adopted to save the animal species shown in the given picture from becoming endangered in India. <i>[Recall]</i>
60.	<i>[Biodiversity]</i> Name the alternate energy obtained from the sources depicted in the given picture and mention <i>one</i> advantage.  <i>[Application]</i>

S.No.	Questions
61.	<i>[Energy]</i> Identify the type of bulb shown in the picture. Analyse why these bulbs are used increasingly in India.  <i>[Analysis]</i>
62.	<i>[Biodiversity]</i> “Overconsumption is an important reason for loss of biodiversity” Give two reasons in support of the above statement. <i>[Analysis]</i>
63.	<i>[Biodiversity]</i> “Paper recycling is the process of turning waste into new paper products.” Highlight two of its uses. <i>[Understanding]</i>

S.No.	Questions
64.	<i>[Managing Soil and Land]</i> Study the following image and answer the questions that follow:  (Source: Shutterstock) (a) Identify the farming technique that is associated with improving the crop yield from the given image. (b) Write <i>one</i> benefit of the above farming technique. [Analysis]
65.	<i>[Managing Soil and Land]</i> What are the key aspects of forest harvesting of NTFPs? [Understanding]

S.No.	Questions
66.	<i>[Wastage]</i> According to the given flowchart, which process is used to treat solid waste? Why is this method suitable?  <pre> graph TD Solid([Solid]) --> Dumping[Dumping] Dumping --> Combustion[Combustion] Combustion --> HeatRecovery[Heat recovery] HeatRecovery --> GasRecovery[Gas Recovery] HeatRecovery --> EnergyProduction[Energy production] GasRecovery --> Emission([Emission from stack]) </pre> <i>[Understanding & Analysis]</i>
67.	<i>[Managing Urban Environment]</i> What is the purpose of land use planning in urban development? <i>[Recall]</i>
68.	<i>[Managing Urban Environment]</i> A display board in a busy square of a city showed ‘Campaign for Clean City’ by the local citizens. This refers to Community participation. How does this help in urban management? <i>[Application]</i>
69.	<i>[Controlling Air Pollution]</i> A developing nation is aiming to reduce its dependence on imported fossil fuels while improving air quality. Evaluate the feasibility and effectiveness of transitioning the public transportation system to CNG in achieving these goals compared to other alternative fuels. <i>[Analysis]</i>
70.	<i>[Controlling Air Pollution]</i> A community living in close proximity to a brick kiln has reported a significant increase in respiratory illnesses among its residents. Analyse the potential health hazards and other environmental issues associated with brick kiln emissions. <i>[Analysis]</i>

S.No.	Questions
71.	<i>[Controlling Air Pollution]</i> “The Bengaluru traffic police aim to maximise data analytics solutions to optimise vehicular movement in the city and reduce congestion.” (Source: Deccan Herald, 13th March 2024) In the context of the given information, what are <i>two</i> possible measures the Bengaluru traffic police can encourage to reduce traffic congestion in the city? [Analysis & Application]



III: Long Answer Questions (5 Marks)

S.No.	Questions
72.	<i>[Environment and Development]</i>  Answer the questions based on the picture above: - Name the type of companies or businesses that operate in more than one nation. (b) Why, according to you, would he want to shift this factory to another country? (c) State any <i>two</i> advantages of spreading out production across the borders. [Application]

S.No.	Questions
73.	<i>[Towards Sustainable Development]</i> Observe the figure carefully and answer the following questions: - Observe the greatest amount of gas from the diagram and name the international agreement which commits its parties to reduce. - Name the environmental agreement ratified by a few governments to reduce the substance shown 14% in the diagram. - Give <i>any two</i> sources for the gas shown 20% in the figure. - How would you add nitrate content in the soil by nature? [Application]
74.	<i>[Addressing Population]</i> <i>Poverty is not only the lack of money but the lack of access to basic human needs like food, shelter, clothing, education, healthcare, etc.</i> Analyse the impacts of poverty that hinder development and growth within a country. [Analysis]
75.	<i>[Energy]</i> <i>Hydropower can protect communities & their livelihood by supplying water that is managed efficiently & safely.</i> Give <i>five</i> ways in which hydropower is advantageous to people & also safeguards the environment. [Application]
76.	<i>[Biodiversity]</i> <i>Man must realise that in order to survive, he must handle nature with care.</i> Based on the statement given above, analyse <i>any five</i> ways in which humans are responsible for the loss of biodiversity. [Understanding & Analysis]
77.	<i>[Biodiversity]</i> <i>Energy is essential for development & sustainable energy consumption is needed for sustainable development.</i> Give <i>five</i> ways in which energy can be used sustainably at home. [Analysis]

S.No.	Questions
78.	<i>[Managing Soil and Land]</i> (a) Which model of community forestry can be initiated for the development of green pastures? (b) What is the purpose of the initiated forestry? [Analysis]
79.	<i>[Wastage]</i> Read the passage and answer the following questions: Vermiculture, also known as worm farming, involves the use of earthworms for composting organic wastes. Earthworms can consume practically all kinds of matter, and they can eat their body weight per day. The compost takes about a month to get ready and contains water-soluble nutrients. (a) Why is vermicomposting important to a farmer? (b) Name two nutrients rich in the excreta (castings) of the worms. [Recall]
80.	<i>[Environment and Development]</i> The large MNCs with worldwide networks look for the cheapest goods in order to maximise their profits. How does this statement impact the developing countries? [Understanding]
81.	<i>[Towards a Sustainable Future]</i> <i>Today, the GEF is an independently operating financial organisation that is working to improve the global environment.</i> Enlist the objective of the Global Environmental Facility that supports the statement above. [Understanding]
82.	<i>[Towards a Sustainable Future]</i> Illustrate the role of NGOs towards Sustainable Development. [Understanding]

S.No.	Questions
83.	<i>[Addressing Population]</i> Shahnaz lives in Luvkushnagar slum in UP with her husband and children, depending on insufficient income from their tailoring services. She had her first child at thirteen. By the age of 28, she had three children, two stillborn twins, increasing hospital bills and debts and declining physical and mental health. (Source: <i>Bayer Global</i>, April 2, 2022) (a) Which strategy do you think this couple should adopt as family planning measures? Mention one. (b) How should the government spread awareness regarding this issue? [Application]
84.	<i>[Energy]</i> SECI (Solar Energy Conservation of India Ltd.) unveils India's largest solar battery project pioneering renewable energy innovation in Chhattisgarh. (Source: <i>PIB</i>, New Delhi, 24th February 2024) (a) Based on the news above, mention <i>any three</i> reasons for promoting solar power in India. (b) State <i>two</i> limitations of solar power which might be challenging for the project. [Analysis]
85.	<i>[Biodiversity]</i> Singapore court gets tough on rhinoceros smuggler. It was Singapore's largest seizure of rhino horns (20 pieces) smuggled via air that transited in the island nation en route from Johannesburg, South Africa, to LAO PDR in 2022. (Source: <i>Traffic.org</i>, 2nd February 2024) (a) Name the convention that banned the trade of rhino horns. (b) Mention the main objective of this convention. (c) Give <i>three</i> points on how harvesting wildlife for commercial purposes can be done sustainably. [Analysis]
86.	<i>[Controlling Air Pollution]</i> A school in a locality of Mumbai made it mandatory to walk or cycle down for the students who stayed in the vicinity and to use only the school bus for students who lived at a distance. This policy was appreciated by the environmentalists. Analyse possible scenarios and reasons why the policy was well-appreciated by environmentalists. [Application & Analysis]
87.	<i>[Managing Urban Environment]</i> List <i>five</i> key factors that are important for town planners to keep in mind while developing a sustainable urban environment. [Recall]

S.No.	Questions
88.	<i>[Managing Urban Environment]</i> Analyse the dangers faced by the people living in the slums. [Recall & Analysis]
89.	<i>[Managing Soil and Land]</i>  (a) What is the best option with regard to the above scenario? (b) State the differences between the two fertilisers. [Understanding]
90.	<i>[Managing Soil and Land]</i> When it comes to global climate change, livestock grazing can be either a blessing or a curse, according to a new study, which offers clues on how to tell the difference. (Source: David L. Chandler MIT News, March 15, 2024.) (a) Highlight two consequences of overgrazing. (b) State <i>three</i> measures to control the menace. [Analysis]

Answer Key

S.No	Expected Answer
1.	(d) all of the above.
2.	(d) None of the above.
3.	(d) sensing by sensors from space satellites.
4.	(a) Both (A) and (R) are true, and (R) is the correct explanation for (A).
5.	(d) Bioremediation
6.	(a) I-B, II-C, III-A, IV-D
7.	(a) Convention on International Trade in Endangered Species (CITES)
8.	(b) Intragenerational
9.	(b) Conversion of forest into agricultural land.
10.	(c) dismantling the caste system.
11.	(d) Gives better yield of crops.
12.	(c) living microorganisms like bacteria and protozoa.
13.	(c) Statement A is correct, and statement B is the reason for A.
14.	(a) Reduces runoff.
15.	(c) A-IV, B-I, C-II, D-III
16.	(b) the enormous increase in population.
17.	(d) meeting the demands of the present generation without compromising the needs of the future generations.
18.	(c) geothermal energy.
19.	(a) habitat centric.
20.	(a) 90% gasoline & 10% ethanol.
21.	(b) captive breeding
22.	(b) CNG is more efficient & less polluting.

S.No	Expected Answer
23.	(b) Statement A is correct & Statement B is the correct explanation of Statement A
24.	(a) A-III, B-I, C-IV, D-II
25.	(b) Global warming
26.	(a) Biogas
27.	(b) increasing agricultural productivity.
28.	(c) reducing wastage of irrigation water.
29.	(a) I and III.
30.	(d) Refuse, Reuse, Recycle, Reduce.
31.	(b) A is true, and R is false.
32.	(d) Development of Women and Children in Rural areas.
33.	(b) over-irrigation of fields.
34.	(d) A-(iii), B-(i), C-(iv), D-(ii)
35.	(c) National Forest Policy, 1988.
36.	(b) to use less water and renewable sources of energy.
37.	(c) Both A and R are true.
38.	(c) Agricultural development
39.	(c) unplanned growth of urban areas.
40.	(d) Statement I is true, and Statement II is the correct reason for I.
41.	b) Redeveloping slums and old buildings to accommodate more houses.
42.	(a) Encouraging carpooling and use of public transport.
43.	(c) X and Z.

S.No	Expected Answer
44.	(any two points) - Food is not distributed equally globally. - Unequal distribution due to differences in soil, climate, political and economic power. - Average income per person varies. - War and corruption, which make it hard for affected people to access food that they or others produced.
45.	World Bank. International Monetary Fund.
46.	(any two points) - Provision of financial and technical assistance. - Granting a period of 10-15 years to adapt and comply with control provisions applicable to developed countries. - Respond quickly to new scientific information and agree to accelerate the required reduction.
47.	The statement given is true. (The percent of the population in age group 0-9 is reduced in figure (b). This shows that the population is reduced marginally.) Reason (any two) : - Proper family planning - Primary health centres - Reduced infant mortality - Improved standard of living - Gender equality - Women empowerment
48.	Yes - It affects India's overall production. - Unemployment in India - Affects the economic development - Use and throw plastic creates pollution in India.
49.	Polyculture and crop rotation.
50.	(any two) - Wetlands have been drained or filled to grow crops or covered with concrete, asphalt and buildings. - Somewhere lost due to mining, logging, oil and gas extraction. - Highway construction and urban development.

S.No	Expected Answer
51.	Biological pest control. Gambusia fish eat mosquito larvae in a pond.
52.	• Non-availability of farmyard manure in sufficient quantity. • Difficulty in growing green manure. • Non-availability of water.
53.	Gene banks are bio-repositories which store genetic material over a long period of time. They are often used for storing the genetic material of species that are endangered or close to extinction. Examples: Pollen bank, Sperm bank.
54.	• They create employment opportunities. • They ensure equitable distribution of income and wealth in the different levels of society.
55.	The image is of a Water Sprinkler used during irrigation. • Saves gallons of water. • Prevents soil erosion.
56.	• Core zone is the legally protected central zone where species are undisturbed by human activities. • Buffer zone outside the core zone is reserved for educational & research activities.
57.	• Critically endangered species are those species which are at very high risk of extinction. Example: Anura. • Vulnerable species are those species which are at high risk of endangerment in the wild. Example: Asiatic Cheetah.
58.	• To conserve biodiversity. • To encourage the sustainable use of the components of biological diversity.
59.	• Project Tiger 1973 scheme had set up biogeographical reserves where core areas were allocated to protect species from human interference & buffer areas were reserved for conservation-oriented programmes. • Habitat restoration and habitat management programmes were also adopted under the Project Tiger scheme.
60.	Biomass energy Advantage: Energy stored in organic compounds can be used to generate electricity
61.	Fluorescent bulbs. These bulbs consume 75% less energy compared to ordinary bulbs, & thus save electricity to promote sustainable energy consumption.

S.No	Expected Answer
62.	- Increasing demands lead to deforestation & thus conversion of forests and wildlife habitats to agricultural & industrial lands. - Overconsumption leads to waste generation & pollution, which causes a loss in genetic diversity.
63.	- Paper can be used to make handmade recycled paper and cardboard. - Recycling paper is used in the packaging industry.
64.	(a) Dry farming (b) <i>(any one)</i> - Suitable for places with low annual precipitation. - Protects the soil from erosion.
65.	<i>(Any two points)</i> - Sustainability. - Diverse product range. - Cultural and traditional knowledge of NTFP harvesting. - Economic value and stability. - Biodiversity conservation. - Commercialisation.
66.	Incineration. This method is suitable for disposal of non-degradable and hazardous wastes from the industries and biomedical wastes.
67.	- The purpose is to allocate the land for different purposes and ensure a sustainable use of resources. - Designing areas and urban spaces that have access to amenities, green spaces, transportation options, and affordable housing.
68.	Community participation: - helps in generating less waste. - creates awareness about keeping surroundings clean.
69.	- Eco-friendly. - Good for the vehicle's engine. - Cheaper than other fuels.

S.No	Expected Answer
70.	• Brick kilns cause air pollution with high levels of dust and smoke. This harms the respiratory system of humans. • The heavy load carried by the workers affects their bones and muscles. • It causes heat stress which harms the health of humans.
71.	• Promote the use of mass transport vehicles. • Existing roads and infrastructure should be improved. • Construction of expressways and flyovers to connected crowded places should be undertaken. • Traffic bottlenecks should be eased.
72.	(a) Multinational Corporation (MNC). (b) Reasons why he might want to shift his factory to another country: • cheaper labour. • availability of natural resources. • easy transport. (c) Advantages of spreading out: • cost of production is low. • earn greater profits. • employment opportunities for the host country. • technology transfer from developed to developing countries.
73.	(a) Kyoto Treaty. (b) The Montreal Protocol. (c) Landfill oil and natural gas systems, agricultural activities, coal mining, mobile combustion, wastewater treatment, natural wetland, paddy rice fields, and emission from livestock production. (d) By growing leguminous and non-leguminous plants like Anabaena, nostoc.

S.No	Expected Answer
74.	(any five) • Child mortality is highest amongst the poor. • Nutritional deficiency. • Lack of proper education. • Many people become homeless. • Lack of access to proper sanitation. • Child labour.
75.	• The construction of dams helps to control floods. • Reservoirs are used for breeding fish. • Ground water reserve is increased by recharging from reservoirs. • It offers aesthetic appeal for tourism. • It is eco-friendly as no waste by-products is produced. (Any other relevant point)
76.	• Overpopulation & thus, overconsumption of resources leads to deforestation. • Deforestation leads to loss of habitat. • Overexploitation of resources leads to pollution & global warming. • Degradation of wetlands for agriculture & industrialisation leads to loss of aquatic species. • The introduction of exotic species & genetically modified organisms in ecosystems threatens the native species.
77.	• Switching off lights & fans when not in use. • Use of fluorescent lamps, which consume 75% less energy than ordinary lamps. • Proper ventilation to allow natural light instead of electricity. • Use of double-glazed windows to insulate the house to reduce the need for air conditioners & room heaters. • Use of thermostats in all electrical appliances to reduce electricity consumption.
78.	(a) Social Forestry. (b) (Any four purposes) • It reduces the pressure on the forests and caters to the needs of the local community for wood and other forest products. • Reduces poverty by providing employment to local people. • Development of cottage industries with the availability of forest products. • Utilization of land according to its productivity • Improvement of ecological condition.

S.No	Expected Answer
79.	(a) Importance: <i>(Any two points)</i> • Vermicompost is an excellent, nutrient-rich organic fertiliser and soil conditioner. • The earthworms increase soil aeration. • The vermicompost improves the holding capacity of the soil. • Enhances germination, plant growth, and crop yield. • Bio wastes conversion reduces waste flow the landfills. (b) Two nutrients rich in excreta of worms: <i>(Any two points)</i> • Phosphorous • Potassium. • Calcium. • Magnesium.
80.	Benefits of MNCs: • Choice of goods in the market rises. • Establishing their manufacturing unit in a developing country, which would be cheap for their production. • Foreign investment by MNCs in these countries has been rising. • Employment Opportunities. Disadvantages of MNCs: • Loss of natural resources. • Pollution. • Worker's health issues.
81.	• To achieve global environmental benefits in the areas of climate change. • Biodiversity • International waters. • Ozone layer depletion. • Promote sustainable agriculture.

S.No	Expected Answer
82.	(Any five points) • Play an important role in mobilising the people at the grassroots. • They need a network among NGOs and an interface between the people and the government to work on community involvement. • Provides information on environmental surveillance and monitoring. • Transmitting development in science and appropriate technology to the people at large. • Generating knowledge regarding traditional and indigenous systems and management practices. • Should be involved in the dissemination of environmental information.
83.	(a) Contraceptives. (b) Measures that the government can undertake for awareness: • Educational campaigns. • The population should be made a part of the school/college curriculum. • By increasing the marriageable age. • The National Literacy Commission and <i>Mahila Samakhya</i> should come forward to spread awareness. • Women should be given proper education.
84.	(a) Reasons for promoting solar energy in India: • Solar energy is infinite & free • It is eco-friendly & free of emissions. • It can be used in remote areas where it is too expensive to extend the electricity power grid. (b) Two limitations of solar power: • Solar energy can be harnessed only during the daytime when it is sunny & ineffective on cloudy days. • Panels & cells don't generate electricity at night.
85.	(a) CITES: Convention on Trade in Endangered Species of Wild Fauna & Flora (b) It is an international agreement between governments to check trade in wildlife so that the existence of any species is not threatened. (c) Sustainable measures for harvesting wildlife for commercial purposes: • Harvesting has to be carefully monitored • The population strength of a species has to be kept in mind while harvesting. • Overharvesting has to be prevented & regeneration time has to be allowed to make sure that the species do not become extinct.

S.No	Expected Answer
86.	• Reduced vehicular air pollution. • Saved fossil fuel. • Eased traffic congestion. • A large number of students living at a distance and therefore coming by bus reduced the number of vehicles travelling to school daily. • There was less honking of horns as individual vehicles were reduced. • As there was less release of greenhouse gases like sulfur dioxide and carbon dioxide, the temperature of the area would be reduced.
87.	• To generate energy from non-polluting resources. • Proper location of sanitary landfills and incineration chambers. • The city should have different zones demarcated for different use (<i>zoning</i>). • The city should have efficient land-use planning. • Transport system: to focus on non-polluting vehicles
88.	• Infections and diseases spread due to poor sanitation. • No security for women. • No formal education facility is provided to children. • Children are forced to work in informal sectors. • Alcoholism is a major problem in slum areas. • Lack of basic civic amenities.
89.	(a) Organic Fertiliser (b) <u>Organic fertilisers</u> • Natural origin: obtained by the decomposition of plant and animal waste. • Does not affect the water bodies. • Does not affect soil fertility in the long run. • Does not affect the groundwater. <u>Chemical fertilisers</u> • Chemical origin: synthesised by the reaction of chemicals. • Enters water bodies and causes eutrophication. • Causes soil salinity and makes the soil unfertile after heavy application. • Contaminates groundwater.

S.No	Expected Answer
90.	(a) Consequences: • Overgrazing reduces the usefulness and productivity of land. • It decreases the biodiversity of the place and leads to soil erosion and desertification. (b) Measures: • The cattle should not be left to graze a particular area over. • Social fencing is the practice of voluntary self-restraint on the grazing of cattle by the people. • Stall feeding.





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